

Solder-Paste Printing and Die-Bonder Placement of Micro-LEDs: Capillary Self-Alignment and Die Tilt

Course report — ET4277 + ET4391 — Part 2

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Abstract—Part 1 described the test board. This part documents the die-bonding process used to populate it with micro-LEDs and analyses the physics that governs the result. After completing the EKL safety training and gaining access to the cleanroom measurement lab, I attended a bonding session run by Ahmed Abdelwahab, who placed Würth WL-SFCC 0404 RGB LEDs on the existing v1 board and trained me on the workflow. The v2 board from Part 1 was still in fabrication, so the session used the v1 board, which carries the same LED footprint. The process is deliberately low-tech: solder paste is printed by hand through a 100 μm stencil, the dies are placed with a Tresky die bonder, and the joint forms on a hot plate. The physics worth studying is in the last step. When the Sn42/Bi57.6/Ag0.4 paste melts, surface tension pulls each die toward the centre of its pad set, which is convenient, but the same forces can tilt a die and lock it in a crooked position that never recovers. The contribution of this report is on the analysis side: it documents the printing and placement steps, then reads the self-alignment and tilt behaviour against the capillary self-alignment literature to identify which process variable matters most. That variable is solder volume. Manual stencil printing gives no volume metering, and volume is exactly the quantity the literature ties most strongly to both placement accuracy and the non-restoring tilt mode.

Index Terms—micro-LED, die bonding, solder paste, stencil printing, capillary self-alignment, die tilt, reflow, cleanroom

I. INTRODUCTION

The bond joint under a micro-LED is the part of the device that the electrical tests in Part 1 were built to characterize. Before any of those measurements can run, the LEDs have to be on the board, and how they get there decides what the joint looks like. This part documents that step and analyses why it behaves the way it does.

The plan with Ahmed was straightforward. Part 1 was the design mini-project. For Part 2, since I am taking ET4277 and ET4391 together, we agreed that he would assemble LEDs on the die bonder while I documented the workflow and wrote a

literature-based analysis of self-alignment and die tilt [1]. The one hard constraint on the board came out of this: it has to fit the die-bonder chuck, so the outline was kept at 9.3 cm $\times$ 9.3 cm, inside the 9.5 cm limit.

Solder bonding is attractive for micro-LEDs because it does three things at once: it mechanically fixes the die, it makes the electrical contact, and, if the geometry is right, it pulls the die into alignment on its own as the solder melts [2]. That last effect, capillary self-alignment, is the reason this is worth studying rather than just doing. It can reach micron-level accuracy with no extra equipment [3], but it is not free of failure modes, and the dominant one for small dies is tilt [4].

II. BACKGROUND: HOW MICRO-LEDs GET ONTO A SUBSTRATE

A micro-LED is a GaN die between roughly 1 and 100 μm on a side, grown on a donor wafer and then moved onto a display or test substrate where it has to be both mechanically held and electrically contacted [5]. Moving millions of them is the step the field calls mass transfer, and it is the recognized bottleneck for micro-LED commercialization [5]. The methods split into two families, and it helps to place where the bonding done in this work sits.

Pick-and-place and its variants. Fine pick-and-place, roll printing, and laser-assisted transfer move dies deterministically, one or a batch at a time. They are accurate but serial, and the placement machine's own accuracy becomes the limit as the die shrinks [6]. The die-bonder flow in this report is the prototyping-scale version of this family: a machine picks each die and sets it on a pad.

Self-assembly. The alternative lets the dies place themselves in parallel under a field, gravity, or, most relevant here, the capillary force of a molten alloy [6]. Cho et al. pushed this to an extreme, assembling 19,663 GaN micro-LEDs of 45 μm diameter at 99.9% yield in under a minute by shaking them over a substrate patterned with molten low-melting-point alloy on gold pads, where the capillary force of the wetting alloy

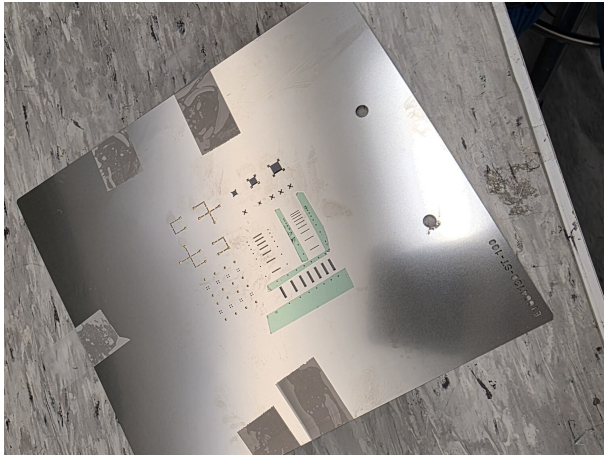


Fig. 1. Manual stencil printing. The $100\ \mu\text{m}$ laser-cut stainless stencil is taped flat over the PCB with green polyimide tape; paste is pushed across the apertures with a plastic spatula and the excess scraped back with a razor blade.

pulls each die onto its site [7]. The throughput is enormous because no machine places the dies.

The two families share the physics this report is about. Even in deterministic placement, once the solder melts the capillary force corrects the position, so the machine only has to get the die close; self-assembly takes that same force and makes it do the whole job. Either way the molten joint aligns the die, and either way the failure mode is the same: the die can tilt rather than settle flat. Studying self-alignment and tilt on a die-bonder-placed LED is therefore directly informative for where the field is heading. The gold (ENIG) pads of the board matter here too, since wetting of the molten alloy to gold is what generates the aligning force [7].

III. CLEANROOM ACCESS AND TRAINING

Access to the EKL cleanroom needed two things first. I completed the EKL safety training and was granted access to the cleanroom and its measurement lab. Ahmed then ran a die-bonding session and trained me on the full flow: how the stencil and paste are handled, how the Tresky die bonder is set up and jogged, how the dies are picked and placed, and how the board is reflowed. The session used Ahmed's existing v1 board (panel E1664138) from the ECTC 2025 work [1], not the v2 board from Part 1, which was still in fabrication at the time. The v1 board carries the same Würth WL-SFCC 0404 footprint, so the process and the pad geometry transfer directly to the v2 board once it arrives. The figures in this report are from that session; the analysis that follows is my own, built from the workflow and the self-alignment literature.

IV. PRINTING AND PLACEMENT

A. Stencil printing

The deposition is manual. The tools are a roll of green polyimide tape, the PCB, a $100\ \mu\text{m}$ laser-cut stainless-steel stencil (the same eC-stencil-mate type Eurocircuits ships with their boards), a plastic spatula, and a razor blade (Fig. 1).

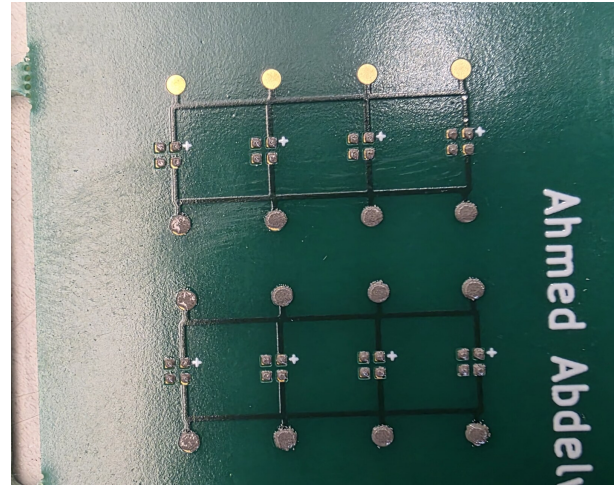


Fig. 2. LED footprints after printing, board still in its Eurocircuits panel. Each four-pad site carries a small paste deposit; deposit size and shape vary noticeably between sites, which is the main limitation of hand printing.

The stencil is aligned over the board and taped down flat, paste is dragged across the apertures with the spatula, and the blade scrapes the excess back. Lifting the stencil leaves a paste deposit on each LED pad.

The paste is Chip Quik TS391LT, a thermally stable no-clean Sn42/Bi57.6/Ag0.4 alloy, T4 mesh ($20\text{--}38\ \mu\text{m}$ particles), 87% metal by weight, with a $138\ ^\circ\text{C}$ melting point [8]. The low melting point is the point: a bismuth alloy reflows around $138\text{--}165\ ^\circ\text{C}$ instead of the $217\ ^\circ\text{C}+$ of SAC305, which keeps the thermal load on the die low.

This method has real advantages. It needs no equipment, it is fast, it is portable, and it works directly with the Eurocircuits stencils and boards. The catch is that the deposit is whatever the operator and the stencil give you: there is no volume metering, the reproducibility is medium, and the smallest aperture that printed cleanly was $0.4 \times 0.4\ \text{mm}^2$, right at the LED pad size. Fig. 2 shows printed deposits across the LED array, and Fig. 3 is a single site under the die-bonder camera. The deposits are usable but visibly uneven from pad to pad.

B. Die placement

Placement runs on a Tresky die bonder (Fig. 4). The board sits on the heated vacuum chuck and the dies wait in a ceramic tray. The machine picks a die, the camera lines it up against the pad fiducial, and it sets the die down on the paste. The placement parameters used were a 100 g place force, a 20 ms place time, and a 40 ms blow-off. These are gentle numbers, which is what you want: the paste has to hold the die by tack until reflow, and pressing too hard squeezes paste out from under the die and starves the joint.

Placement accuracy by hand-jogged die bonder is only moderate. Fig. 5 shows a die just after it was set down: it sits low and slightly rotated relative to the crosshair that marks the pad centre. This offset is normal and, within limits, fine, because the next step is supposed to correct it.



Fig. 3. One LED footprint at high magnification through the die-bonder camera. The four 0.4 mm pads each hold a grainy paste deposit; the cross is a placement fiducial. The dark cross between the pads is the soldermask web that separates the anode and the three cathodes.

C. Paste and reflow

The choice of paste shapes the whole alignment step, because the die only moves while the solder is liquid. The TS391LT is a near-eutectic Sn42/Bi57.6/Ag0.4 alloy with a 138 °C melting point, well below the 217 °C of SAC305. For a GaN micro-LED this low budget matters: less heat into the die means less thermal stress on the epitaxy and on the nascent joint. The trade-off is that bismuth alloys are more brittle than SAC, so the joint is mechanically weaker, which is acceptable here because the boards are for characterization rather than field service.

Reflow was done on a hot plate toward a peak near 165 °C. The window between melting at 138 °C and the peak is the time the joint is liquid and therefore the time available for self-alignment to act. This is the part of the process least under control in a manual flow: a hot plate heats from the bottom and the board's thermal mass and contact set the real profile, so the liquidus time varies between runs. Since the capillary restoring motion needs a finite time to bring the die home before the solder freezes, an inconsistent profile turns directly into inconsistent alignment, on top of the volume variation from hand printing.

V. SELF-ALIGNMENT AND DIE TILT

A. Why the die moves on its own

When the paste melts, the joint becomes a liquid bridge between the die and its pads. Surface tension tries to minimize the surface energy of that bridge, which means it pulls the die toward the position where the molten solder columns are shortest and most symmetric, i.e. centred on the pads [2], [9]. For a molten solder the surface tension is around 500 mN/m and the capillary length is about 2 mm [9], so for a sub-millimetre die the capillary force easily dominates gravity. The

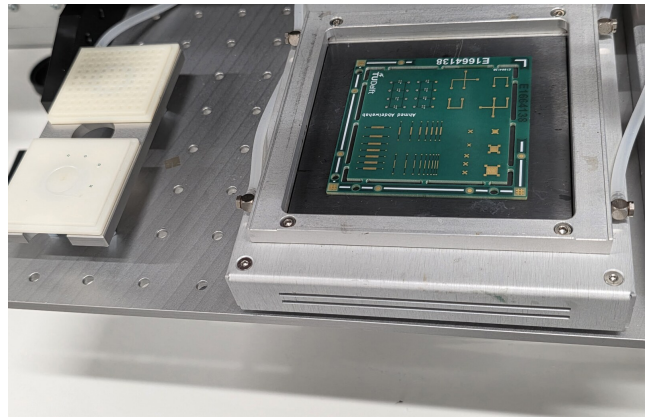


Fig. 4. The board mounted on the Tresky die bonder's heated chuck, with the LED dies in the white ceramic tray at left. The optical breadboard underneath isolates the placement head from vibration.

restoring force scales with the in-plane offset, so a die placed slightly off-centre (Fig. 5) gets dragged back during reflow. This is the mechanism that makes the moderate placement accuracy acceptable.

How much offset the joint can absorb is not small. Krammer et al. modelled the chip during reflow with four forces — surface tension, hydrostatic and capillary pressure, gravity, and dynamic friction — and showed by experiment that surface tension dominates the in-plane motion [10]. In their measurements, surface-mount chips self-aligned even from initial misplacements of 400–500 μm [10], with average correction distances around 249 μm along one axis [11]. They also found the alignment is anisotropic and geometry-dependent: parts with metallized sidewalls (so the solder can wet a vertical face) self-align noticeably better than parts without [11]. That is encouraging for the WL-SFCC, whose four bottom contacts give the molten joint defined features to pull against, and it sets a generous tolerance budget against the moderate die-bonder placement of Fig. 5.

The recent work of Lesueur et al. puts numbers on the accuracy ceiling. Using mass reflow of corner solder pillars they reached lateral alignment better than 2 μm , and in some configurations under 1 μm , with measured z-tilt variation also below 1 μm , and they showed this self-alignment absorbs standard flip-chip pick-and-place placement tolerances outright [12]. The lesson for the manual process here is that the achievable accuracy is set by the joint geometry and the solder, not by how carefully the die is placed.

B. The tilt problem

A floating die has four ways to move: shift, twist, lift, and tilt. Three of them are stable, meaning the capillary force restores the die toward alignment if it is perturbed. Tilt is the exception: it is non-restoring [4]. Once a die starts to rock, the solder geometry does not push it flat again, and for a small die with a thin solder layer the low corner can touch the pad and pin the die in a tilted position that reflow never undoes. Berthier et al. show exactly this with an LED

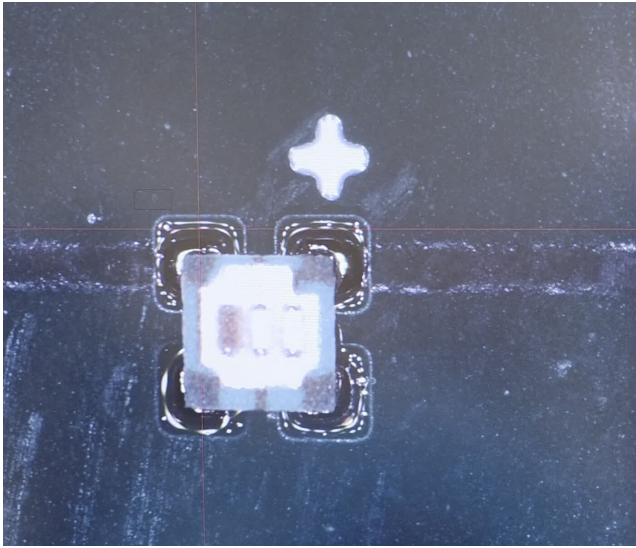


Fig. 5. A WL-SFCC die set down on the printed paste, seen through the die-bonder camera. The die straddles the four pads but sits offset and rotated relative to the crosshair (pad centre) and the cross fiducial above it. Capillary forces are expected to pull this offset out during reflow; the risk is that the same forces tilt the die instead.

on solder, and note that avoiding it otherwise requires near-perfect horizontality at drop-off, which is hard to guarantee on a hand-jogged machine [4]. The older reflow-dynamics models capture why: the Ellis-Masada picture has the chip rotate about a corner that stays pinned to the pad [11], so any asymmetry in the solder under the four corners feeds straight into a tilt the joint cannot correct.

The variable that couples placement, alignment, and tilt is the solder volume. Vareilles et al. measured this directly for sub-millimetre cells and found the best placement accuracy with low solder volume, a receptor pad smaller than the die, and an initial offset of about 10% of the die size; too much solder both shifts and tilts the part [3]. That is the uncomfortable conclusion for the process I ran: hand stencil printing has no volume control, and volume is the one knob that matters most. The uneven deposits in Figs. 2 and 3 are the visible form of that problem.

VI. ANALYSIS AND SUGGESTIONS

Reading the process against the literature, and discussing it with Ahmed, points to four conclusions:

The print is the weak link, not the placement. The die bonder places to moderate accuracy and the capillary force forgives the rest, so chasing tighter placement is not where the effort should go. The deposit volume and its pad-to-pad consistency are what set how well, and how flat, each die ends up. A finer or volume-controlled deposition (a thinner stencil for these pads, or jetting the paste) would attack the actual limiting variable.

Keep the solder volume low and the pad inside the die footprint. Following Vareilles et al., a pad slightly smaller than the die and a modest deposit should give a stronger, flatter pull than a generous deposit on a matched pad [3]. The current

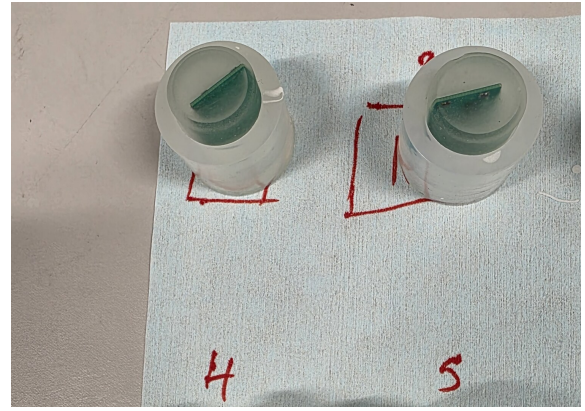


Fig. 6. Diced board coupons with bonded LEDs, sealed in labelled vials after the session. Cutting the board into per-site coupons lets each bonded LED be cross-sectioned or imaged from the side, which is the view that actually reveals tilt.

0.4 mm pads sit right at the die contact size, so there is room to trim them.

Tilt is the failure to watch, not shift. Because shift and twist self-correct and tilt does not, inspection should look for rocked dies specifically, ideally in cross-section or with side-view imaging rather than top-down, which hides tilt. If tilt turns out to be common, the lyophilic-band trick from Berthier et al. — patterning the wettable area to constrain the liquid and stabilize the tilt mode — is the design-side fix [4], and it would go straight onto a future revision of the board.

Mind the reflow profile. The TS391LT melts at 138 °C and we reflowed on a hot plate toward a peak near 165 °C. Hot-plate reflow heats from the bottom and the temperature is not as controlled as an oven profile, so the time the joint spends liquid, which is the time available for self-alignment, varies. A logged profile would make the alignment window repeatable.

VII. CONCLUSION

WL-SFCC micro-LEDs were bonded in the EKL cleanroom in a session run by Ahmed, which I attended after completing the EKL safety training, gaining cleanroom and measurement-lab access, and being trained on the die-bonding flow. The process — hand stencil printing of a low-melting Sn/Bi/Ag paste, die-bonder placement, hot-plate reflow — is cheap and fast, and the capillary self-alignment of the molten joint covers the moderate placement accuracy. The analysis here identifies solder-volume control as the limiting variable: the literature ties volume to both alignment accuracy and the non-restoring tilt mode, and hand printing does not control it. The concrete next steps that follow from this are tighter deposit control, pads trimmed inside the die footprint, tilt-aware inspection, and a logged reflow profile. The session was run on the v1 board; once the v2 boards from Part 1 arrive from the fab, the same process and these same conclusions apply directly, and the bonded v2 boards are what the electrical characterization will run on. The samples from this session were diced and stored for inspection (Fig. 6); side-view imaging of those coupons is

the natural next check, since it is the one view that shows tilt directly rather than hiding it.

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